# MedSync Resuscitation Desk — context brief for Claude

Sep 23, 2026 · @Someone

## How to read this brief

This brief explains MedSync's resuscitation desk to Claude in a chat that cannot see the code. It was written from the repository (branch `feat/booking-app`) and its decision records (ADRs), as of the date above.

**MedSync** is a role-based digital-hospital platform for Ugandan hospitals, starting with internal medicine at Mulago. It is built with Next.js 16, Supabase Postgres and row-level security. Separate workspaces serve reception, nurses, doctors, hospital admins, patients and a super-admin. The clinical path runs triage → resus → encounter (clerkship) → prescribing.

When you answer questions about the desk:

- **Decision support only.** The desk records what clinicians do and suggests the next step. It never orders, acts or administers anything itself.
- **Never supply a dose from memory.** Every dose on the desk traces to a cited, verified source and a Clinical Safety Officer (CSO) sign-off. If a dose is not stated here, say so.
- **Paediatric doses are deferred.** The desk never shows an adult number for a child.
- **Stay within this brief.** If something is not covered here, say it is not documented rather than guessing.

## What the resuscitation desk is

The resuscitation desk is a full-screen bedside workspace. It guides a team through the ABCDE primary survey (Airway, Breathing, Circulation, Disability, Exposure) for a critically ill patient. As the team works, it writes down every assessment, action, drug and vital sign on a permanent, tamper-evident record.

- **Who it is for:** a Red / Level-1 (L1) patient from triage, a patient with a Step-0 life-threat, or anyone who deteriorates in resus.

- **Who uses it:** any active clinical staff member at the hospital. Nurses usually open it and doctors usually lead. Reception staff (non-clinical cadres) are blocked, both in the app and in the database.
- **Where it lives:** nurses use `/nurse/resus` (the board) and `/nurse/resus/[episodeId]` (the desk). Doctors have a mirror at `/doctor/resus/...` that reuses the same desk component. The desk opens full-bleed, with the workspace sidebar hidden.
- **Security:** you must be signed in with multi-factor authentication (AAL2). Access is scoped to your hospital. Every time a patient's name is revealed, the reveal is logged.
- **One resus = one episode.** An episode has a mutable header (active or concluded, disposition, outcome, arrest time). It also has an append-only timeline of events.

The desk sits in the clinical flow like this:

```
flowchart LR
  T[Triage] -->|Red / L1 or<br/>Step-0 life-threat| B[Resus board]
  B -->|Open resus| D[Resus desk<br/>ABCDE]
  D -->|arrest detected| X[ALS module<br/>to follow]
  D -->|Conclude| O[Disposition<br/>+ outcome]
  O --> E[Encounter / ward]
```

The principle underneath: **the app records, the clinician leads.** Tapping anything logs it as done by the named clinician. The desk suggests the next step based on guidelines, but it never makes a clinical decision or blocks the team.

## Getting a patient onto the desk

There are four ways to open a resus. All of them start the permanent record, summon the team and redirect to the desk.

| Entry point | Where | What happens |
|---|---|---|
| **Rapid Red** | Triage worklist, ⚡ button | One tap opens a small confirm dialog. Pick one of 7 life-threats: airway compromise/stridor, absent or agonal breathing, no pulse/arrest, unresponsive, active seizure, exsanguinating haemorrhage, critically low $SpO_2$. The patient becomes Level-1 with no full triage form. It can only over-triage, never under-triage, and it is audited. |
| **Open resus** (from triage) | Triage worklist row already at Red/L1 | Opens or resumes the desk using that patient's most recent L1 triage record. |
| **Open resus** (from the board) | Resus board, "Red / L1 — open a resus" list | Lists L1 triage records from the last 24 hours that have no episode yet (up to 50). Each row shows age, the life-threat reason and time since triage. |
| **Start ad-hoc resus** | Resus board footer | Opens an unlinked episode for a deterioration that happens in resus, with no triage record behind it. |

**What opening does:**

1. **Opening is idempotent.** If an active episode already exists for that triage record, you are sent to it instead of a duplicate.
2. **The trigger kind is recorded:** `step0_life_threat` (fast-tracked in triage), `triage_red`, or `in_resus_deterioration` (ad-hoc).
3. **Triage vitals become the first resus reading.** They are stamped with the triage time, so the vitals trend starts where measurement did.
4. **Team mobilisation fires.** See "Team mobilisation and roles" below. A failed alert never rolls back the episode.

**The resus board** (`/nurse/resus`, `/doctor/resus`) is the shared starting point:

- **"Active resuscitations"** lists running episodes, newest first. A pulsing red dot shows when any are running. An episode with a detected arrest is highlighted and labelled "⚠ ARREST · CPR / ALS".

- **"Red / L1 — open a resus"** lists the waiting L1 patients described above.
- **A footer hint** tells doctors and clinical officers that their cadre is recommended to lead. It tells everyone else that any responder can take any role.
- **Patient names** are decrypted only after the scope check, and each reveal is audited. A patient with no identity shows as "Unidentified patient".

## The desk layout

The desk is a single full-screen page with five zones, stacked top to bottom. On wide screens the centre and the right rail sit side by side. Closed or cancelled episodes stay readable, but every control is disabled.

**1. Sticky command header**

- **Board back-link,** a red `RESUS` tag, the patient's name (or "Unidentified patient"), age and sex.
- **Trigger reason,** hospital and department.
- **A live elapsed clock** (m:ss or h:mm:ss) with a pulsing dot while the episode is active. When the episode ends, a state badge shows the disposition instead.
- **The ABCDE step bar:** five buttons, one per letter. Each shows its state (see "Visual language"). The step you are on has a teal focus ring. The recommended next step has a dashed teal outline.
- **"Return to A"** jumps back to Airway and logs "Returned to A (reassess from the top)".
- **"⚠ Arrest"** declares cardiac arrest in one tap. It is hidden once an arrest has been declared.

**2. Arrest banner.** After an arrest is declared, a full-width red banner reads "⚠ CARDIAC ARREST — Start CPR". Below it: "Routed to: ALS cardiac-arrest module — to follow".

**3. Vitals strip** (continuous monitoring)

| Element | What it shows |
|---|---|
| **Vital cells** | HR, RR, $SpO_2$, SBP (with DBP inside), temperature, glucose, $EtCO_2$. Each shows the current value, "was …" (the previous value) and a trend arrow with the size of the change. |
| **Derived cells** | Shock index (2 decimal places) and mean arterial pressure (MAP), with the same trend display. |

| Element | What it shows |
|---|---|
| **Flag chips** | Pulse pressure (flagged when narrow relative to SBP) and qSOFA (flagged at ⩾ 2). These have no trend graph. |
| **Trend graph** | Hover (desktop) or tap (touch) a cell. It shows a time-aware line with the normal band shaded, dashed danger thresholds and the latest point enlarged. BP plots SBP and DBP together. It needs at least 2 readings. |
| **Record vitals** | A capture row: HR, RR, $SpO_2$, SBP, DBP, temperature, glucose, GCS, $EtCO_2$, urine output, and ACVPU (Alert, new Confusion, Voice, Pain, Unresponsive). |
| **Re-check nudge** | After any drug, fluid, procedure or action newer than the last vitals, the button turns red and reads "↻ Re-check vitals" (reassess after every intervention). |

Cell colour follows the trend: rose when deteriorating, teal when improving, otherwise the severity band. A critical value always stays red, even while improving. Arrow direction shows the raw movement. Arrow colour shows clinical meaning, so a rising $SpO_2$ is good and a falling SBP is bad.

**4. Centre column**

- **An "Open life-threat(s)" alert** appears when any step has an open threat: "… — treat before proceeding".
- **The Anaphylaxis bar** is always present, because anaphylaxis applies at every step (A, B, C and E). It shows the IM adrenaline dose line, plus a Detail toggle (recognition, adjuncts, what is not recommended, refractory route). A **Prescribe / record** button opens the prescribing dialog.
- **The step panel** for the chosen letter: Airway, Breathing, Circulation, Disability or Exposure. See the next section.

**5. Right rail**

- **Team panel:** roles, claim or take a role, the "I'm responding" button and the count responding.

- **Resus record:** the live append-only timeline, newest first. Each entry shows the bedside clock time, a glyph by kind, the `[step]` tag, the label and any dose. Entries are tinted by state (failed or danger = faint rose, performed = faint teal, in progress = faint blue). A "saving…" row appears while a write is in flight.

After every write made from this screen, the desk re-reads the server. Entries made by other team members appear on your next action or page reload. There is no live push.

## The ABCDE engine, step by step

All five steps share one pattern: **findings → inferred state (and cause) → triggered actions → a state for each action.** Each step's rules are a pure function covered by tests (40–60 tests per step). The clinical content is reference data transcribed from a CSO-signed clinical catalog, and every action cites its guideline source.

### What every step panel has

- **Context toggles** at the top (for example Trauma, Allergen trigger, No pulse), then finding chips grouped by Look / Listen / Feel / Measure.
- **A "Record … findings" button.** It saves the finding-set as an `assessment` event. After an intervention it changes to "↻ Re-record … findings".
- **A state indicator.** The engine classifies the step using a "worst finding wins" rule: an optimistic finding never hides a dangerous one.
- **The indicated-actions board.** It is a card that stays beside the findings on wide screens. Each action shows finding → why → do: the trigger, the rationale and the guideline source.
- **Six action states,** each set with one tap: `indicated`, `in_progress`, `performed`, `not_indicated`, `deferred`, `failed`. Marking an action **failed** shows its next-tier escalation inline.
- **Drug actions** open the prescribing dialog pre-filled. See "Emergency prescribing".
- **Treat before proceeding.** A "proceed guard" warns while the step is still threatened. It informs and never blocks, because a resus is non-linear. An unassessed step is never treated as safe.
- **Arrest routing.** Signs of arrest in any step route to the arrest seam (see below).
- **No schema change.** Findings, actions and states are all timeline events. The current picture is recomputed from the timeline on every render.

## A — Airway (the template for B–E)

- **Findings:** patency, Look/Listen/Feel signs, and the difficult-airway screen (LEMON / MOANS / RODS mnemonics). Context: Trauma/C-spine, Allergen trigger, Gag/clenched jaw, No signs of life.
- **States:** secure → threatened → partially obstructed → completely obstructed → apnoea-ventilate → apnoeic-arrest.
- **Actions:** Tier 1/2/3, least invasive first, including the DAS 2025 Plan A→B→C→D ladder (a failed plan escalates to the next).
- **Safety rules:** apnoea with a pulse means bag-mask ventilation, not CPR. See-saw breathing counts as complete obstruction. Suspected epiglottitis never triggers instrumentation. Trauma means jaw thrust, and airway patency outranks strict C-spine immobilisation. IM adrenaline for angioedema requires anaphylaxis features.

## B — Breathing

- **Findings:** Look, Listen, Feel & percuss, and banded Measure codes (RR, $SpO_2$, $PaCO_2$, PEF, $EtCO_2$ thresholds). Context: Chest trauma, Allergen trigger, Known asthma, COPD/$CO_2$-retainer risk, Suspected PE, No signs of life.
- **States:** adequate → distress-compensating → failure-decompensating → peri-arrest → apnoeic.
- **Clusters it recognises:** tension pneumothorax, massive haemothorax, open chest wound, acute or life-threatening asthma, COPD, pulmonary oedema, pneumonia/sepsis, PE and anaphylaxis.
- **Pre-arrest traps classed as peri-arrest:** silent chest, falling RR or exhaustion, a normal or rising $PaCO_2$ in asthma, and rising $EtCO_2$ with a falling RR. Normal $SpO_2$ with high work of breathing still flags. Cyanosis is labelled a late sign. In anaphylaxis, adrenaline is surfaced before salbutamol.

## C — Circulation

- **Findings:** Look, Feel, and banded Measure codes (SBP/MAP, shock index, pulse pressure, lactate, ECG, β-hCG, ultrasound). Context: Trauma, Active bleeding, Febrile child (FEAST), Pregnancy possible, Spinal injury, Allergen trigger, Head injury/TBI, No pulse.
- **Two outputs.** The state runs well perfused → compensated → decompensated → peri-arrest → arrest. The **shock type** is hypovolaemic (haemorrhagic or not), distributive (septic, anaphylactic, neurogenic), cardiogenic, obstructive (tamponade, PE, tension)

or undifferentiated.

- **Actions depend on the shock type.** Crystalloid is never offered for cardiogenic, obstructive or haemorrhagic shock. Haemorrhage means stop the bleeding, blood / major-haemorrhage protocol, TXA and permissive hypotension (with a TBI exception at MAP ⩾ 80).
- **Other traps:** a normal BP never downgrades shock. A febrile child's crystalloid carries the FEAST no-flooding caution. Shock in a woman of reproductive age prompts β-hCG (ectopic). Obstructive shock means relieve the obstruction. An undifferentiated picture prompts bedside ultrasound.

## D — Disability

- **Findings:** GCS component pickers (Eye /4, Verbal /5 with "T" when intubated, Motor /6), AVPU, pupils, focal signs, seizure, posture, glucose and raised-ICP signs. Context: Opioids suspected, Low RR, Pregnant/postpartum, Head injury/TBI, Alcohol/malnourished, Fever + meningism, A/B/C abnormal, No pulse.
- **Two outputs.** The state runs alert → mild-moderate → severe/coma (GCS ⩽ 8) → active seizure/status → raised ICP/herniation. The **neuro-syndrome** is hypoglycaemia, opioid, status epilepticus, eclampsia, raised ICP, stroke, meningitis, DKA/HHS, non-convulsive, secondary to A/B/C, or undifferentiated.
- **What sets D apart:** GCS total plus a serial trend. A falling GCS triggers a return to A.
- **Traps:** "Don't Ever Forget Glucose" pins *check glucose* at the very top and flags a missing glucose. Reduced GCS is treated as an A/B/C problem first. GCS ⩽ 8 routes back to A for a definitive airway. Cushing's response means raised ICP, so don't treat the BP. Give benzodiazepines at the full guideline dose. Eclampsia gets magnesium first, not a benzodiazepine. Pinpoint pupils with reduced consciousness suggest opioids, so naloxone.

## E — Exposure (closes the loop)

- **Findings:** Skin/rash, Bleeding/wounds/injury (log-roll the back), Patches/lines, core temperature with staging, and an AMPLE history (Allergies, Medications, Past history, Last meal, Events). Context: Trauma (log-roll), Environmental exposure, Serotonergic/neuroleptic, Left exposed (re-cover!), No pulse.
- **Two outputs.** The state runs no new critical finding → significant → temperature extreme → critical. The **suspected diagnosis** is anaphylaxis, meningococcal sepsis, occult haemorrhage, necrotising fasciitis, limb threat, severe hypothermia, hyperthermia, hypothermia or opioid patch.

- **Actions work as a checklist.** Expose, measure temperature, re-cover, AMPLE and return to A are always shown. Every critical finding also adds its action unconditionally, so one diagnosis never hides another.
- **Traps:** a non-blanching rash means antibiotics without delay. Unexplained hypothermia prompts a septic screen ("cold sepsis"). Occult bleeding means log-roll and go back to C. Re-cover and keep the patient warm. A transdermal opioid patch means remove it and give naloxone. Temperature staging rounds to the more severe band.
- **AMPLE allergies feed prescribing.** Allergies captured here are stored where the prescribing allergy check reads them.
- **"Return to A → reassess"** auto-saves the draft (so AMPLE allergies are never lost), logs the loop event and moves the desk to A. If the step isn't finished, the button reads "Return to A anyway".

## Cardiac arrest and the return-to-A loop

The desk **detects** cardiac arrest and hands off. It deliberately does **not** run the cardiac-arrest algorithm (ALS, advanced life support).

**How it detects arrest (RCUK/ERC 2025).** Arrest means **unresponsive AND (not breathing normally and/or no central pulse / no signs of life)**. Agonal gasps count as not breathing normally. Each sign can be yes, no or unknown, and too few signs means no arrest. The check is available at every letter:

- the **"⚠ Arrest"** button in the header (a one-tap declaration: the clinician is asserting the criterion)
- a **No pulse** or **No signs of life** toggle inside any step panel

**When arrest is detected:**

1. The episode is stamped with `arrest_detected_at`. The timeline gets `arrest_detected` and `cpr_started` events, and the change is audited.
2. A red banner fills the desk: "Start CPR now — begin chest compressions, call the resus team, get the defibrillator."
3. The route is labelled **"ALS cardiac-arrest module — to follow"**. A hypothermic arrest adds the "not dead until warm and dead" modifier.
4. On the board, the episode is highlighted as "⚠ ARREST · CPR / ALS".

**Not built, on purpose:** rhythm assessment, defibrillation energies, CPR rate/depth/ratio, **any arrest drug dosing** (adrenaline in arrest, amiodarone), the 4 Hs & 4 Ts, and post-ROSC care. These belong to a separate, separately-verified ALS module that does not exist yet. Until then the team continues on paper. Peri-arrest arrhythmias are flagged but not built.

**Apnoea with a pulse is not arrest.** In A and B it routes to *open the airway + bag-mask ventilation*, not CPR. A safety review caught this mis-routing in Airway before sign-off.

**Return to A.** A resus is a loop, not a single pass. You can go back to A three ways:

- **"Return to A"** in the header, any time. It logs "Returned to A (reassess from the top)".
- **The end of E** ("Return to A → reassess"). It auto-saves the E draft and logs the loop event.
- **Engine triggers:** GCS ⩽ 8 or a falling GCS in D, and an opioid picture (naloxone, then back to B).

The loop is always advisory. An urgent jump back to A is never blocked.

## Team mobilisation and roles

Opening a resus summons the team straight away, following the chain recognition → call for help → response (ERC 2025). Roles are recorded but never block anyone.

**Who gets summoned.** Every active clinical staff member in the receiving department, plus every active ward in-charge at the hospital. There is no on-shift model yet, so this list is deliberately wide and may over-notify.

**How they are summoned:**

- **In-app notification (live today).** One `sa_notifications` row per recipient, kind `clinical_resus_mobilisation`, subject "RESUS — …". The body reads: "A resuscitation has been activated (…) in your department. Converge now. Open the resus board for the live view."
- **SMS / push (stubbed).** Calls go through a stub that only logs the intent. Swapping in a real gateway is a one-file change.
- **One `resus_mobilisation_alerts` row per recipient** tracks who is responding.
- **It never throws.** A failed alert never rolls back the episode: the record is the source of truth, and the alert is a courtesy.

**The Team panel** (right rail of the desk):

- **Seven roles:** Leader, Airway, Circulation, Scribe, Runner, CPR, Family (family liaison). Each shows its holder, "You", or "—".
- **"Claim"** an empty role, or **"Take"** one someone else holds. Taking a role closes the previous holder's assignment (a hand-over) and opens yours. Every change is audited and logged on the timeline as `role_change`.
- **Leader is recommended, never required,** for doctor and clinical-officer cadres (intern, medical officer, consultant, clinical officer). A ★ marks it. Any responder can take any role under pressure.
- **"I'm responding"** acknowledges your alert. The header shows "N/M responding", then "✓ You're responding — claim a role above".

The code also has a server action to assign a role to *another* person, but the desk UI currently offers only claim/take for yourself.

## Emergency prescribing from the desk

In a resus, drugs are grabbed from the crash cart and given at the bedside, so **recording the administration is the prescription**. Every drug action on the desk (step actions and the Anaphylaxis bar) opens one **prescribe dialog**. It is pre-filled from a verified emergency drug catalog and checked by a pure safety engine.

**The drug catalog** is cited reference data. It was cross-checked on the web on 2026-06-03 (36 of 36 entries verified) against RCUK/ERC/AHA 2025, GINA 2025, BTS-SIGN, JBDS and BNF/SmPCs. Each entry holds:

- **The adult dose details:** dose, concentration, route, interval and maximum.
- **A paediatric note** (weight-based).
- **Safety data:** cautions, contraindications and monitoring, a controlled-drug flag, and a look-alike guard.
- **Its source and a confidence tag.**

Concentration is recorded, not just mg. Category-level items (GTN, furosemide, vasopressor infusions) show no fixed dose. Rapid-sequence-intubation (RSI) drugs are record-only.

**One tap from an action to a filled-in prescription.** A recommendation code such as `c_txa` or `d_glucose` resolves to one or more pre-filled options (for example hydrocortisone or prednisolone). Each option carries its own safety flags for this patient.

**Six safety checks run at prescribe time.** Each returns block, warn or info:

| Check | Behaviour |
|---|---|
| Allergy | Blocks on a match (substring match, so cross-reactivity is not covered) |
| Concentration look-alike | Guards drugs with more than one strength, e.g. adrenaline 1:1000 IM vs 1:10 000 IV |
| Paediatric routing | Age < 16 never gets an adult flat dose. Weight is required or the check blocks. |
| Max / cumulative | Sums doses across the episode, e.g. atropine 3 mg |
| Contraindication | Blocks on a matching condition, e.g. adenosine + asthma |
| Controlled drug | Requires a witness, even in an emergency |

**Editing a dose.** An "Adjust" expander lets the clinician edit dose, route, concentration, frequency, duration, volume, rate, PRN and notes. All checks **re-run live on the edited values**, and the guideline default stays visible, struck through. A plausibility guard compares the edited dose to the catalog:

- A normal adjustment (roughly 0.5–2×) raises nothing.
- Further out warns, then blocks.
- An egregious dose (≥ 5× or ≤ 0.2×, or ≥ 3× the top of a ranged dose) needs the dose typed in to confirm.

**Blocks can be overridden,** because a resus is time-critical. Overriding needs a recorded reason, and both the override and its flags are audited.

**Allergies:**

- **The dialog will not let you forget them.** If none are recorded, it says "Allergies not yet recorded — check before giving", with a capture field and an NKDA (no known drug allergies) button.
- **Where the check reads from:** allergies captured in this resus (the latest capture wins; NKDA clears), AMPLE allergies from step E, and clerked allergies from the encounter record ( `patient_allergies` , read only when the encounter engine is live).

**Who may prescribe:**

- **Authority comes from the clinician's cadre** (routine vs controlled prescribing).

- **A non-prescriber is never blocked.** The dialog says: "You are not an authorised prescriber for this drug — recording as administered under the resus protocol. A prescriber should authorise / witness."
- **Co-sign.** Where a co-sign is required, the prescriber is recorded as witness.

**What gets recorded:** a `drug` event on the permanent timeline. It holds the drug, the dose actually given, concentration, route, who gave it, time, patient, cart, amount in mg, controlled flag, witness, override reason and flags, source, the guideline default, and a field-level edit log (field, from, to). This is also the future pharmacy module's reconciliation record: restocking and expiry. The pharmacy module itself is not built.

**Fixed adult doses in the catalog** (quoted from the app's verified reference; paediatric doses are banded or deferred):

| Indication | Encoded adult dose |
|---|---|
| Anaphylaxis | Adrenaline 500 µg IM, repeat after 5 min. Routine corticosteroids not recommended. |
| Hypoglycaemia | 75 mL 20% or 150 mL 10% glucose IV; glucagon 1 mg IM; oral 15–20 g |
| Oxygen (critically ill) | 15 L/min; target $SpO_2$ 94–98% (88–92% if at risk of $CO_2$ retention) |
| Sepsis | Crystalloid ≥ 30 mL/kg (weak recommendation); target MAP ≥ 65 (strong) |
| Acute asthma | Salbutamol 5 mg neb; ipratropium 500 µg neb; prednisolone 40–50 mg PO or hydrocortisone 100 mg IV; IV magnesium 1.2–2 g over 20 min |
| Status epilepticus | Benzodiazepine at full dose: lorazepam 4 mg or midazolam 10 mg |
| Eclampsia | Magnesium 4 g IV over 5–15 min, then 1 g/h for 24 h; recurrent seizure 2–4 g over 5 min |

Category-level items show no number, and the clinician doses them: naloxone in some contexts, nebulised adrenaline for airway oedema, PE thrombolysis, GTN, furosemide, vasopressors. TXA pins only the time-critical fact; the dose comes from the local major-haemorrhage protocol.

## The record: append-only, audited, reconstructable

The resus timeline is the medico-legal record for the debrief and any later review. It cannot be silently edited or deleted, and it can be replayed exactly as it stood at any past moment.

**Two parts:**

- **`resus_episodes`** is the mutable header: state, trigger, disposition, outcome, arrest time, and links to the triage record, intake, patient and encounter. Every change to it is audited.
- **`resus_timeline_events`** is hard append-only. A database trigger rejects any UPDATE or DELETE, and those privileges are revoked from every client role.

**Event kinds (14):** assessment, action, drug, fluid, procedure, vital, decision, role_change, mobilisation, reassessment, arrest_detected, cpr_started, handover, note. Each event is tagged with a step (pre, A–E or arrest) and records who logged it.

**Two clocks.** `recorded_at` is bedside time, when it happened. `created_at` is system time, when the row was written. Triage vitals carried into a resus keep their triage time.

**Corrections append, never overwrite.** A correction is a new event pointing at the one it replaces ( `supersedes_event_id` ). The original stays forever and is only hidden from the active view. `reconstructTimeline(events, {asOf})` is pure and tested. It builds the active view and can replay the record at any past instant.

**Privacy and audit:**

- Free-text narrative is encrypted at rest (AES-256-GCM).
- Every write also lands a row in `sa_audit_log` . That covers opening an episode, each event, role assignments, arrest, conclusion and rapid-red triage.
- Name reveals on the board and the desk are logged to `clinical_phi_access_log` .

**Vitals** are recorded on the timeline. They are also copied to `clinical_observations` , the patient's long-term observations record.

**How a resus ends.** An episode ends as `concluded` or `cancelled` , stamped with an end time. It records a disposition (theatre, ICU, cath lab, ward, mortuary, transferred or deceased) and a free-text outcome. After that the desk is read-only and the header shows the state and disposition.

**Built, but no button yet.** The server actions `concludeResus` and `correctResusEvent` exist, but **no button in the current desk calls them**. See "Clinical safety posture and known gaps".

**Access (row-level security).** Reading an episode is limited to:

- a super-admin
- the patient
- whoever recorded the entry
- active clinical staff at the episode's hospital

All writes go through the service role inside audited server actions, never directly from the client.

## Visual language

On the desk, **colour means state, and red is scarce.** Red appears only for a genuine life-threat now or a failed action. Everything else uses a calm ramp: teal for good or done, amber for open or recommended, grey for pending or parked. The aim is that under pressure the loudest thing on screen is the letter that is killing the patient now.

**Step-bar states:**

| State | When | Look |
|---|---|---|
| Pending | Not yet assessed | Grey |
| Clear | Assessed, no threat | Light teal |
| Addressed | Threat treated | Deep teal, filled letter chip |
| Threatened | Open threat, not life-threatening now | Amber |
| Active threat | Life-threat now | Filled red, white chip |

What counts as a life-threat now, by step:

- **A:** complete obstruction or apnoea
- **B:** peri-arrest or apnoea, or a top-priority emergency such as tension pneumothorax or life-threatening asthma
- **C:** decompensated shock, peri-arrest or arrest, or haemorrhage control
- **D:** raised ICP, status epilepticus, GCS ⩽ 8 or a falling GCS

- **E:** a critical finding or a temperature extreme
- **Any step:** a detected arrest

Some prompts are always pinned, such as *check glucose* in D and *measure temperature* in E. These never turn a letter red on their own.

**Action-state tokens:**

| State | Colour | Icon |
|---|---|---|
| Indicated | Amber | CircleDot |
| In progress | Blue, gentle pulse only if motion is allowed | LoaderCircle |
| Performed | Teal | Check |
| Failed | Red, shows the next-tier escalation | TriangleAlert |
| Deferred | Grey | Pause |
| Not indicated | Outline only | Minus |

**Urgency ordering.** One shared function sorts every action board:

- First by state band: failed → indicated → in progress → deferred → performed → not indicated.
- Then by how time-critical the engine rates the action.
- Ties keep their original order.

So each board reads must-escalate → recommended → underway → parked → done → N/A. The sort **never removes** an action; done and N/A items drop to the bottom but stay visible. Each board header shows "N outstanding" (failed + indicated + in progress). A polite screen-reader live region announces the most urgent action.

**Standing rules:**

- **Never colour alone.** Every state also has a word, and actions also have an icon. Trend arrows pair colour with a glyph and a value.
- **Accessible contrast.** WCAG 2.2 AA in light and dark, with a fallback that still reads in greyscale.
- **No left-edge accent bars.** State is shown by full border, tinted fill, ring, chip, icon and word.

- **Findings stay calmer than actions.** A selected finding uses the teal selected style; the step's state indicator carries the urgency.
- **Restrained motion.** Active-threat never flashes. Pulses run only when the user allows motion. Only the newest timeline entry animates in, and the trend graph appears without animation.
- **Headings are weight 500, not bold.** This follows MedSync's design system ( `DESIGN-medsync.md` ).

## Data model and code map

The resus desk is four database tables, one server-actions file, about twenty domain modules and 332 unit tests. It is built in Next.js 16 (App Router, Server Actions), Drizzle ORM and Supabase Postgres.

**Tables** (migration `0038_resus_abcde.sql`, schema `db/schema/resus.ts`):

| Table | Holds |
|---|---|
| `resus_episodes` | hospital, department, triage record, intake, patient, encounter, trigger kind and reason, state, disposition, outcome, started/ended/arrest timestamps, who opened it |
| `resus_timeline_events` | episode, kind, step, recorded by, recorded_at, created_at, JSON payload, encrypted narrative, supersedes_event_id (hard append-only) |
| `resus_role_assignments` | episode, user, role, assigned_at, ended_at (open = current holder), assigned by |
| `resus_mobilisation_alerts` | episode, recipient, channel, linked notification, sent/delivered/acknowledged, responding |

Related: `emergency_administrations.resus_episode_id` and `patient_allergies` (migration 0039, smart prescribing).

**Routes and UI:**

- `app/nurse/resus/page.tsx` + `_components/resus-board.tsx` : the board
- `app/nurse/resus/[episodeId]/page.tsx` + `_components/resus-desk.tsx` : the desk (about 1,580 lines)

- `_components/{airway,breathing,circulation,disability,exposure}-panel.tsx`: the five step panels
- `_components/prescribe-dialog.tsx`, `resus-actions-heading.tsx`
- `app/doctor/resus/**`: the doctor mirror, reusing the nurse components
- Triage entry points: `app/nurse/triage/_components/open-resus-button.tsx` and `rapid-red-button.tsx`

**Server actions** (`app/nurse/resus/actions.ts`):

| Group | Actions |
|---|---|
| Open | openResusFromTriage, openResusFromIntake, openBlankResus |
| Events | logResusEvent, correctResusEvent, logResusVitals |
| Per step | record{Airway,Breathing,Circulation,Disability,Exposure}Findings, set{…}ActionState, recordResusReturnToA |
| Drugs | recordResusAllergies, recordEmergencyAdministration, loadPersistentAllergiesForEpisode |
| Team | assignResusRoleAction, claimResusRoleAction, acknowledgeResusAlertAction |
| Arrest / end | detectResusArrestAction, concludeResus |

**Domain logic** (`lib/resus/`):

- **Engines:** `engine.ts` (tracker, next step, proceed guard), plus `airway.ts`, `breathing.ts`, `circulation.ts`, `disability.ts`, `exposure.ts` and `arrest.ts`.
- **Support:** `action-order.ts`, `vitals.ts`, `vital-trend.ts`, `timeline.ts`, `record.ts`, `mobilise.ts`, `board.ts`, `scope.ts`, `types.ts`.
- **Reference data** (`reference/`): `abcde-catalog.ts`, one catalog per step, and `doses.ts`.
- **Prescribing:** `lib/prescribing/engine.ts`, `administration.ts`, `allergy-merge.ts` and `reference/emergency-drug-catalog.ts`.

**Tests** (`tests/resus/`, 332 tests):

| File | Tests |
|---|---|
| airway | 60 |
| circulation | 57 |
| breathing | 54 |
| disability | 49 |
| exposure | 40 |
| engine | 17 |
| vital-trend | 17 |
| vitals | 10 |
| action-order | 9 |
| timeline | 9 |
| arrest | 5 |
| doses | 5 |

**Decision records:**

| ADR | Topic |
|---|---|
| 0044 | Record immutability |
| 0045 | ABCDE engine and doses |
| 0046 | Team mobilisation |
| 0047 | Arrest seam |
| 0048 | Airway |
| 0049 | State colour system |
| 0050 | Breathing |
| 0051 | Smart prescribing: catalog and engine |
| 0052 | Prescribing accountability and pharmacy seam |

| ADR | Topic |
|---|---|
| 0053 | Circulation |
| 0054 | Editable prescription parameters |
| 0055 | Disability |
| 0056 | Exposure |
| 0057 | Urgency ordering |
| 0058 | Vitals trend |

First-stage design proposals ("GATE-A") live in `docs/resus-*-gate-a-proposal.md`. The CSO clinical catalogs live in `docs/clinical/resus-*-clinical-catalog.md` and `emergency-crash-cart-drug-catalog.md`.

## Clinical safety posture and known gaps

The desk is **decision support with a CSO-signed clinical model**. Its gaps are either deliberate, quarantined scope or UI wiring that has not been done yet.

**Safety posture:**

- **Signed off.** The CSO signed the first-stage clinical design for every step: the ABCDE model, doses and team roster on 2026-06-02 (Airway included); Circulation, Disability and Exposure on 2026-06-03.
- **Reviewed.** Each step went through an adversarial safety review before sign-off. The Airway review fixed 5 clinical-safety issues.
- **Verified doses.** Every fixed dose is web-verified and cited, and tests enforce the "never a remembered dose" rule. Flumazenil is marked *needs CSO verification* by design. The plausibility band widths for edited doses are for the CSO to ratify.
- **Never blocks.** The app never acts, never auto-administers and never hard-blocks the team. Overrides are recorded and audited.

**Deliberately not built** (these stay in the record and prompt escalation, and are flagged as deferred):

- **The cardiac-arrest ALS module:** rhythms, defibrillation, arrest drugs, 4 Hs & 4 Ts, post-ROSC care.

- **Procedural tooling:** full RSI checklist, ventilator/NIV titration, chest drain, pericardiocentesis, REBOA, the full major-haemorrhage workflow, vasopressor titration, the hypothermia/ECLS pathway, heatstroke cooling, burns (TBSA/Parkland), dantrolene, secondary survey, toxidromes.
- **Paediatric and obstetric depth.** The engine routes by age but never shows an adult dose to a child. Vital-sign bands use adult cutoffs.
- **Pharmacy reconciliation.** The data is captured; the pharmacy module is not built.
- **Out-of-band alerts.** SMS and push are stubbed; only in-app alerts are live. There is no on-shift model, so everyone in the department is notified.

**Built, but not yet usable from the desk** (worth knowing if asked "how do I …"):

| Gap | Effect |
|---|---|
| No **Conclude / Cancel** button ( `concludeResus` has no UI) | Episodes stay "active" on the board; disposition and outcome can't be set from the desk |
| No **correct an entry** control ( `correctResusEvent` has no UI) | A mis-logged drug or action can't be superseded from the desk. Re-recording findings does replace the step's picture. |
| No **handover** (SBAR/RSVP) or **free-text note** entry | Those event kinds exist but nothing writes them from the UI |
| No **assign a role to someone else** (only claim/take for yourself) | The leader can't allocate roles on a responder's behalf |
| **No live sync** between devices | Others' entries appear on your next action or reload |
| `resus_episodes.encounter_id` is never set | No automatic hand-off from the resus to an encounter record |

**Other limits:**

- Allergy matching is by substring, with no cross-reactivity check. Clerked allergies are read only once the encounter engine flag is live.
- DBP and $EtCO_2$ have no danger band, so their trend always shows neutral.
- There are no row-level-security tests for the four resus tables in `tests/rls/`.

- The five panels each keep their own action card (known duplication, deferred cleanup).

**When answering about the desk:** describe it as clinician-led decision support. Point to the deferred modules rather than improvising them, and never fill a dose gap from memory.